Balancing Motion Sickness Mitigation and Immersion in Viewing 360 Videos Using Camera Trajectory Visualization

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Figure 1: In the Clouds: Vancouver from Cypress Mountain.

ABSTRACT

360° videos provide an accessible form of immersive media for virtual reality (VR), yet their characteristic camera motion can induce motion sickness during viewing. Existing mitigation techniques often introduce visual distractions or distortions, potentially compromising immersion. Drawing on the theory of controllability in motion sickness, this paper investigates whether visualizing the capturing camera's motion trajectory can mitigate motion sickness while preserving the immersive experience. We designed two interfaces that visualize camera trajectories in 360° VR: an overlay interface, which presents the trajectory within the viewer's field of view, and a minimap interface, which provides a spatial overview of the trajectory. Across two user studies, we compared these interfaces with three conditions: unmodified 360° VR, a counter-optical-flow visualization, and a dynamic field-of-view technique. Results indicate that both trajectory-based interfaces effectively reduced motion sickness while preserving users' sense of immersion. Our work demonstrates the potential of camera-trajectory visualization as a less intrusive approach to balancing motion sickness mitigation and immersion in 360° VR.

Index Terms: Radiosity, global illumination, constant time.

1 INTRODUCTION

A coherent introduction could be structured into **6 paragraphs**, moving from the broad problem to the specific gap, theoretical grounding, proposed approach, and research contribution:

* **Paragraph 1 — 360° VR as accessible immersive media**

* Introduce 360° videos as an increasingly accessible form of immersive content for VR. * Explain their advantages: users can freely explore the surrounding visual environment and experience recorded scenes from an immersive first-person perspective. * Highlight that unlike interactive VR experiences, 360° videos preserve real-world visual content while requiring relatively limited production and interaction infrastructure. * Transition to a key challenge: the camera motion embedded in 360° videos can cause motion sickness.

* **Paragraph 2 — Motion sickness in 360° video viewing**

* Explain why camera motion in 360° videos can induce motion sickness, particularly when visually perceived motion is not accompanied by corresponding physical motion. * Briefly discuss symptoms and their negative effects on comfort, viewing duration, and the overall VR experience. * Emphasize that mitigating motion sickness is therefore important for making 360° VR more comfortable and usable. * Establish the tension between reducing sickness and preserving the immersive qualities that motivate users to watch 360° content in VR.

* **Paragraph 3 — Existing mitigation approaches and their limitations**

* Review existing visual approaches for reducing motion sickness in VR/360° video. * Organize them around techniques such as **visual distraction**, **visual distortion**, **counter-optical-flow cues**, and **dynamic field-of-view manipulation**. * Explain that although these approaches can reduce users' exposure to provocative visual motion, they may obscure, distort, or restrict the visual scene. * Argue that such interventions can consequently reduce **visual immersion**, creating a trade-off between motion sickness mitigation and the quality of the immersive experience. * Identify the need for approaches that mitigate sickness **without substantially altering the visual content**.

* **Paragraph 4 — Controllability as a theoretical basis**

* Introduce the **theory of controllability in motion sickness** as an alternative perspective for mitigation. * Explain that perceived

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controllability over visually induced motion can influence users' susceptibility to motion sickness. * In prerecorded 360° videos, viewers cannot control the physical motion of the capturing camera, making the camera's movement potentially unpredictable or difficult to anticipate. * Motivate the idea that explicitly communicating the **camera's motion trajectory** could provide users with information about upcoming or ongoing movement, thereby increasing their perceived controllability. * Position trajectory visualization as a way to address motion sickness through **additional information rather than visual distortion**.

* **Paragraph 5 — Proposed approach and research questions**

* Introduce the proposed approach: visualizing the **capturing camera's motion trajectory** during 360° video viewing. * Describe the two interface designs:

* **Overlay:** directly presents the camera trajectory within the user's view. * **Minimap:** presents the trajectory as a spatial overview, allowing users to understand camera movement from a broader perspective. * Explain the design rationale and how the two interfaces provide trajectory information through different spatial representations. * State that two user studies evaluate these interfaces against:

* a baseline 360° VR condition without visual enhancement, * a counter-optical-flow visualization, * and a dynamic field-of-view technique. * Frame the investigation around two central questions: **(1) Can camera-trajectory visualization reduce motion sickness? (2) Can it do so while preserving users' sense of immersion?**

* **Paragraph 6 — Contributions**

* Summarize the paper's contributions in **2–3 concrete points**:

* **A design approach** for mitigating motion sickness in 360° VR through visualization of the capturing camera's motion trajectory, grounded in controllability theory. * **Two interface designs**, overlay and minimap, that operationalize trajectory visualization in different spatial representations. * **Empirical evidence from two user studies** comparing trajectory-based interfaces with existing visual mitigation approaches, demonstrating their ability to balance motion sickness mitigation and immersion. * Conclude by positioning the work as contributing to the design of **less intrusive motion-sickness mitigation techniques for immersive 360° video viewing**.

2 Related work

3 Interfaces Leveraging Camera Trajectory Visualizations

4 Study 1

Study 1 examined how different visual interventions affected visually induced motion sickness (VIMS), presence, and subjective preferences while participants viewed moving-camera 360° videos in VR. We used three approximately one-minute videos depicting walking, cycling, and driving. Each video was presented under four conditions: an unmodified *Baseline*, *Dynamic FOV*, our *Minimap* trajectory visualization, and our *Overlay Path* trajectory visualization. Thus, the study followed a 4 (Method) × 3 (Video) within-subjects design. Study 2 included *Reverse Optical Flow* for direct comparison with our two trajectory-visualization techniques.

4.1 Participants

We recruited 36 participants. All participants completed all 12 experimental conditions (four methods × three videos), yielding 432 observations with no missing quantitative data. To reduce carryover of VIMS symptoms from one condition to the next, participants rested for 15 minutes between consecutive conditions; after every third condition, the washout period was extended to 30 minutes.

4.2 Study Design and Procedure

Each participant viewed the walking, cycling, and driving videos under all four visualization conditions. Each exposure lasted approximately one minute and was followed by the study measures described below and the scheduled washout period. To reduce order effects, we counterbalanced the presentation order of the four visualization conditions using a balanced Latin square.

4.3 Measures

Simulator sickness. We assessed sickness using the 16-item Simulator Sickness Questionnaire (SSQ) and computed the Kennedy weighted scores for Total Severity (TSC), Nausea (N), Oculomotor discomfort (O), and Disorientation (D). Higher scores indicate more severe sickness symptoms. Besides, to ensure every participant understand the 16 questions correctly, we use the Vietnamese version of the original 16-item SSQ.

Immersion. We also assessed the participants' sense of immersion using a five-item, 7-point Likert questionnaire adapted from established measures of immersion in virtual environments. Responses ranged from 1 (strongly disagree) to 7 (strongly agree), with higher mean scores indicating stronger perceived immersion.

Preference ranking and post-study interviews. After completing all 12 conditions, participants took part in a semi-structured interview about their VIMS experience and the three mitigation techniques. Questions focused on perceived sickness reduction, anticipation of camera motion, visual stability, attention demands, and trade-offs between comfort and presence. Participants also discussed specific usability issues, such as looking away from the scene to check the minimap, excessive attention drawn by the overlay path, and the restricted peripheral view caused by Dynamic FOV.

A subset of 13 participants (36.1%) additionally ranked the four conditions from 1 (most preferred) to 4 (least preferred). Because rankings were available for only 13 of 36 participants, we treated them as exploratory descriptive evidence. Interview responses were analyzed thematically to help interpret differences between the mitigation techniques.

4.4 Analysis

For the SSQ scores and presence ratings, we conducted two-way repeated-measures ANOVAs with *Method* (Baseline, Dynamic FOV, Minimap, Overlay Path) and *Video* (Cycling, Walking, Driving) as within-subject factors. Greenhouse–Geisser corrections were applied to the repeated-measures F tests. For the four SSQ outcomes, Holm corrections were applied across dependent variables within each effect family, and Bonferroni-adjusted pairwise comparisons were used for significant Method effects. We report generalized eta squared (η^2_G) as the effect size.

The SSQ residuals deviated from normality, primarily because of heavy tails rather than pronounced skewness. We therefore assessed robustness by repeating the SSQ analyses after excluding the three participants associated with extreme outliers. The significance pattern was unchanged across all combinations of SSQ effects and dependent variables, so the results below use the full sample ($N = 36$).

4.5 Results

4.5.1 VIMS and SSQ

We conducted a two-way repeated-measures ANOVA with Method and Video as within-subject factors. As summarized in Table 1, **Method significantly affected all four SSQ measures**, whereas neither Video nor the Method × Video interaction was significant. This indicates that the mitigation effects were broadly consistent across the walking, cycling, and driving videos.

Table 1: Repeated-measures ANOVA results for Study 1. Greenhouse–Geisser-corrected degrees of freedom are reported. Holm correction was applied across the four SSQ outcomes.

Effect	TSC	Nausea	Oculomotor	Disorientation	Presence
Method	$F = 9.02$, $p < .001$	$F = 7.37$, $p < .001$	$F = 8.63$, $p < .001$	$F = 6.98$, $p < .001$	$F = 13.64$, $p < .001$
Video	$p = .230$	$p = .229$	$p = .084$	$p = .410$	$p = .450$
Method × Video	$p = .535$	$p = .570$	$p = .392$	$p = .642$	$p = .109$

Table 2 summarizes the main differences between visualization methods. Lower SSQ scores indicate less VIMS, while higher Presence scores indicate a stronger sense of presence.

Dynamic FOV. Dynamic FOV produced the **largest numerical reduction in VIMS**, reducing TSC by approximately 33% relative to Baseline. However, it did not significantly outperform Minimap or Overlay Path on any SSQ measure.

Minimap. Minimap reduced TSC by approximately 22% while maintaining the **highest Presence score among the three mitigation methods**.

Overlay Path. Overlay Path reduced TSC by approximately 21%. Its reductions relative to Baseline were significant for TSC, Nausea, and Oculomotor discomfort, while the reduction in Disorientation did not reach significance after Bonferroni correction ($p = .0734$).

4.5.2 Presence and the VIMS–Presence Trade-off

The pattern for Presence differed from the SSQ results. Table 3 highlights the pairwise comparisons most relevant to the VIMS–Presence trade-off.

Dynamic FOV. Although Dynamic FOV achieved the largest numerical VIMS reduction, it also produced the **lowest Presence score** ($M = 3.92$), significantly below the other three conditions.

Minimap. Minimap preserved Presence most effectively: its score did not significantly differ from Baseline ($p = .1984$).

Overlay Path. Overlay Path produced a smaller but significant reduction in Presence relative to Baseline ($p = .0093$).

These results indicate a clear **VIMS–Presence trade-off**: stronger sickness mitigation does not necessarily correspond to a better overall viewing experience.

4.5.3 Qualitative Feedback

The interviews further explain the differences between the three mitigation methods. Table 4 summarizes the main qualitative observations together with the corresponding quantitative results.

Overall, the interviews suggest that the effectiveness of each technique depends not only on sickness reduction, but also on how the visual aid affects attention, and even the exposure duration to the 360 video: **some participants even suggest that for longer 360 videos, they would trade immersion with VIMS reduction**.

4.5.4 Exploratory Preference Ranking

Because preference rankings were available for only 13 of the 36 participants (36.1%), these results are treated as exploratory. However in study 2, a complete reference ranking is available.

Overlay Path. Overlay Path had the **best overall mean rank and Borda score** and was almost never ranked last.

Dynamic FOV. Dynamic FOV again showed the strongest polarization: it received both the **largest number of first-choice rankings** and the **largest number of last-choice rankings**.

4.5.5 Study 1 Summary

Taken together, Study 1 shows that all three visualization techniques reduced VIMS relative to the unmodified Baseline, while their effects remained consistent across the three video types. **Dynamic FOV** provided the greatest numerical sickness reduction but substantially reduced Presence and produced polarized responses. **Minimap** provided the strongest quantitative balance between VIMS mitigation and Presence preservation. **Overlay Path** achieved a similar VIMS reduction while receiving the strongest exploratory preference ranking, although interview feedback highlighted the importance of clear and visually stable implementation.

5 Study 2

Study 1 showed that both trajectory visualizations reduced VIMS relative to unmodified 360° video while costing less Presence than Dynamic FOV. However, Study 1 compared them with only one other mitigation technique and measured sickness only after each trial. Study 2 therefore compared our *Minimap* and *Overlay Path* interfaces with *Reverse Optical Flow* (ROF), a counter-optical-flow visualization. We treated Minimap and Overlay Path as the *trajectory-based* interfaces and ROF as the *non-trajectory* comparison, and asked: **are trajectory-based interfaces as effective as ROF at reducing VIMS, while being less intrusive?** Study 2 also measured sickness both before and after every trial, and collected a complete preference ranking and interview from every participant.

5.1 Participants

We recruited 36 participants. Nineteen had used a VR headset before and 17 had not. All participants completed all nine experimental conditions (three methods × three videos), yielding 324 trials and 648 SSQ administrations with no missing data.

5.2 Study Design and Procedure

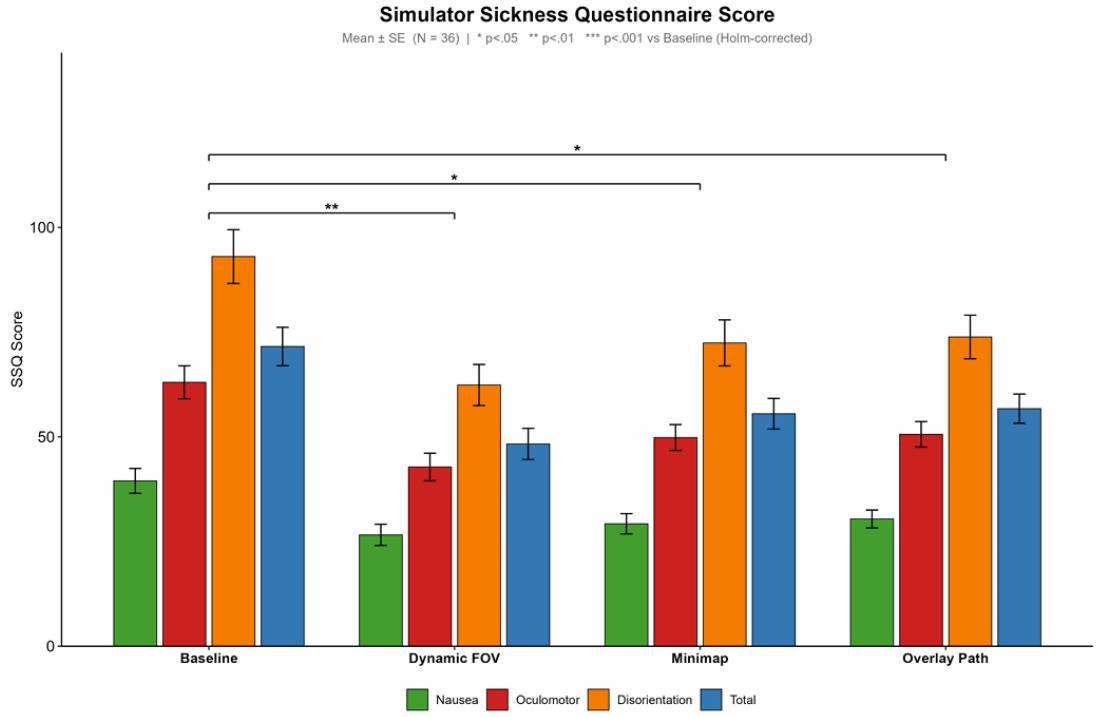
Study 2 followed a 3 (*Method*: ROF, Minimap, Overlay Path) × 3 (*Video*: Walking, Cycling, Driving) within-subjects design. Unlike Study 1, it had **no Baseline condition**, so all comparisons are between mitigation techniques. Each participant experienced the three methods in separate blocks, and each block contained all three videos. Method order was fully counterbalanced: each of the six possible orders was assigned to six participants. Video order within blocks was counterbalanced so that each video appeared equally often in each position. Participants completed the SSQ immediately before each video, and the SSQ and the Presence questionnaire immediately after it. The median interval between finishing one trial and starting the next was 6.3 minutes (IQR 5.1–7.3). After all nine trials, participants ranked the three methods and took part in a short interview.

5.3 Measures

Simulator sickness. We used the same Vietnamese version of the 16-item SSQ and the same Kennedy weighted scores (TSC, N, O, D) as in Study 1, but administered the questionnaire both before and after every trial. Our main outcome was the change score $\Delta = \text{post} - \text{pre}$, where positive values mean that symptoms increased during viewing. The pre-trial scores also let us check that participants started each condition at a similar level of sickness.

Table 2: Main quantitative results by visualization method, collapsed across video type. SSQ and Presence values are means ± SD. Percentage reductions in TSC are relative to Baseline.

Method	TSC	TSC reduction	Nausea	Disorientation	Presence
Baseline	71.6 ± 47.6	–	39.5 ± 30.8	93.1 ± 66.7	4.79 ± 0.94
Dynamic FOV	48.3 ± 38.7	**33%**	26.6 ± 26.4	62.4 ± 50.9	3.92 ± 1.33
Minimap	55.5 ± 38.0	22%	29.2 ± 25.2	72.4 ± 57.1	4.53 ± 1.00
Overlay Path	56.7 ± 36.3	21%	30.4 ± 22.1	73.9 ± 54.0	4.41 ± 1.02



Hình: SSQ grouped bar chart theo Method với significance brackets (Holm-corrected). Error bars = ±1 SE.

Figure 2: Simulator Sickness Questionnaire scores across visualization methods. Lower scores indicate less VIMS. Error bars show ±1 SE; brackets indicate significant pairwise comparisons.

Table 3: Key Bonferroni-adjusted pairwise comparisons for Presence.

Comparison	Δ Presence	p
Baseline – Dynamic FOV	+0.861	**.0006**
Baseline – Minimap	+0.254	.1984
Baseline – Overlay Path	+0.372	**.0093**
Dynamic FOV – Minimap	-0.607	**.0022**
Dynamic FOV – Overlay Path	-0.489	**.0101**
Minimap – Overlay Path	+0.119	1.000

Immersion. Participants completed the same five-item, 7-point presence questionnaire as in Study 1 after every trial (Cronbach's $\alpha = .78$).

Preference ranking and post-study interviews. After all nine trials, every participant ranked the three methods from 1 (best) to 3 by how well each balanced immersion and VIMS mitigation, and explained the ranking in a semi-structured interview. Later participants were also asked which method best reduced their sickness (21 named one method) and which felt most realistic (26 named one method). We coded the interview notes with a codebook of eleven codes in three families: *perceived sickness*, *perceived realism*, and *interface experience* (occlusion, distraction, glance cost, route anticipation, timing, visual noise, visibility, familiarity, and agency). Each code was counted at most once per participant and method, so all counts below are numbers of participants.

To address intrusiveness, we counted a *recorded intrusiveness-related complaint* when a participant criticized a method for blocking the scene (occlusion), drawing attention away from it (distraction), requiring gaze shifts (glance cost), or producing flicker or visual clutter (visual noise). Participants were not asked directly whether a method was intrusive, so this measure counts unprompted complaints; a method without complaints is not thereby shown to be unobtrusive.

5.4 Analysis

For the SSQ, we conducted 2 (*Time*: pre, post) × 3 (*Method*) × 3 (*Video*) repeated-measures ANOVAs. The *Time* × *Method* interaction is the effect of interest, because it tests whether the methods differed in how much sickness increased during viewing. As in Study 1, we applied Greenhouse–Geisser corrections, Holm correc-

Table 4: Integration of quantitative and qualitative findings.

Method	Quantitative result	Interview feedback
Minimap	TSC reduced by 22%; Presence remained closest to Baseline ($p = .1984$).	Familiar and stable for some participants, but its small size and low position required users to look away from the main scene, producing additional attention cost.
Overlay Path	TSC reduced by 21%; small Presence reduction relative to Baseline ($p = .0093$).	Helped some participants anticipate the camera trajectory. However, visual distraction and difficulty initially understanding the function of the path could interfere with the experience.
Dynamic FOV	**Largest TSC reduction (33%)**, but **lowest Presence** ($M = 3.92$) and highest Presence variability ($SD = 1.33$).	Highly polarized. Some participants reported that reducing peripheral motion helped alleviate discomfort, while others disliked the restricted peripheral view.

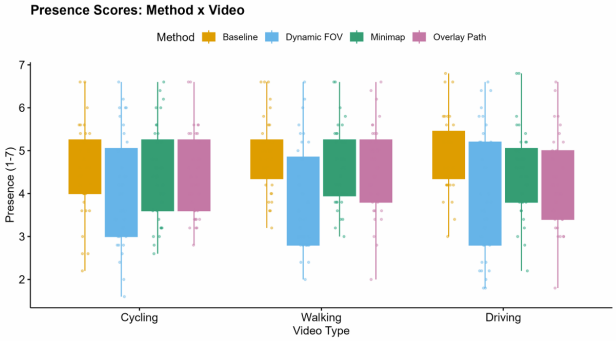


Figure 3: Presence ratings across visualization methods and video types. Higher values indicate stronger perceived presence.

Table 5: Exploratory preference ranking. Lower mean rank indicates stronger preference. First- and last-choice counts are tie-adjusted.

Method	Mean rank	Borda	First	Last
Overlay Path	**2.04**	**38.5**	4.0	**0.5**
Minimap	2.42	33.5	1.0	1.0
Dynamic FOV	2.73	29.5	**5.0**	**6.5**
Baseline	2.81	28.5	3.0	5.0

tions across the four SSQ outcomes, and report η^2_G. Presence was analyzed with a 3×3 repeated-measures ANOVA.

To compare the trajectory-based interfaces with ROF, we computed for each participant the difference between their mean ΔTSC under Minimap and Overlay Path and their ΔTSC under ROF (contrast weights $-1, \frac{1}{2}, \frac{1}{2}$). Because the question is whether the trajectory-based interfaces are *as effective as* ROF, we used a **non-inferiority test** (one-sided $\alpha = .05$) with a margin of $+5$ TSC points: the trajectory-based interfaces are non-inferior if the upper limit of the 90% confidence interval of the difference lies below $+5$. The margin is one third of the smallest reduction relative to Baseline in Study 1 (14.9 TSC points, Overlay Path), so a non-inferior method keeps at least two-thirds of that benefit. We checked the result with a mixed-effects model on all 324 trials, a participant bootstrap, and separate tests for each trajectory-based interface.

For intrusiveness, we computed for each participant the difference between whether they complained about ROF (0 or 1) and the mean of the same indicator for Minimap and Overlay Path. We tested whether this difference was positive with an exact permutation test that enumerates every assignment of the three method labels within each participant. Answering the research question in the affirmative required both the non-inferiority test and the intrusiveness test to be significant; because both must succeed, no further α adjustment is needed.

For the ranking, we used a Friedman test with Kendall's W, and a Plackett–Luce model in which Minimap and Overlay Path share one preference parameter.

The SSQ change scores were zero-inflated and heavy-tailed. Rank-transformed ANOVAs, Friedman tests, bootstrap confidence intervals, and a mixed model excluding extreme values all led to the same conclusions, so we report the parametric results below.

5.5 Results

5.5.1 VIMS and SSQ

Table 6 summarizes the ANOVA results and Table 7 the scores for each method. **Viewing increased sickness under every method**: *Time* was significant for all four SSQ outcomes, and TSC rose from 9.55 before to 17.98 after viewing (+8.44, 95% CI [5.27, 11.61], $d_z = 0.90$). **The method did not affect how much sickness increased**: the *Time* × *Method* interaction was not significant for any SSQ outcome (all $F \leq 0.78$, uncorrected $p \geq .44$, $\eta^2_G \leq .001$). As Figure 4 shows, TSC increased by 9.00 points with ROF, 9.07 with Minimap, and 7.24 with Overlay Path. Participants started each method at a similar level (pre-trial TSC 9.90, 9.38, and 9.35; all $p \geq .77$), so these differences reflect what happened during viewing.

The **video type** mattered more than the method. The *Time* × *Video* interaction was significant for TSC, Oculomotor discomfort, and Disorientation: the Walking video increased TSC by 13.09 points, roughly 2.4 times the increase for Driving (5.44) and nearly twice that for Cycling (6.79). Walking differed from both other videos (Bonferroni $p \leq .003$), whereas Cycling and Driving did not differ. The three-way interaction was not significant, so no method was better suited to a particular video.

Symptoms partly carried over between trials: the post-trial TSC of one trial predicted the pre-trial TSC of the next ($b = 0.39$, $p < .001$). Because method order was fully counterbalanced, this carryover adds noise but does not favor any method.

5.5.2 Are Trajectory-Based Interfaces as Effective as ROF?

Table 8 and Figure 5 show the non-inferiority results. **The trajectory-based interfaces were non-inferior to ROF in measured VIMS.** On average, TSC increased 0.85 points *less* with the trajectory-based interfaces than with ROF (90% CI $[-2.54, 0.84]$). The upper limit was well below the $+5$ margin ($p < .001$), and the result held in the mixed-effects model (upper limit 2.41) and the bootstrap (0.76). **Both interfaces were individually non-inferior**: the upper limit was 2.71 for Minimap and 0.69 for Overlay Path (Holm-corrected $p \leq .002$). The difference also lay within ± 5 points in a two-sided equivalence test, so the trajectory-based interfaces were not meaningfully *better* than ROF either. No SSQ subscale differed between the groups (all Holm-corrected $p \geq .83$), and the contrast did not vary by video ($F(1.73, 60.71) = 0.82$, $p = .43$).

Because Study 2 has no Baseline, this result shows that the trajectory-based interfaces were about as effective as ROF; it does not show that any of the three methods reduced VIMS relative to unmodified 360° video.

Table 6: Repeated-measures ANOVA results for Study 2. SSQ outcomes were analyzed with a 2 (Time) $\times$ 3 (Method) $\times$ 3 (Video) ANOVA; Presence, measured after viewing only, with a 3 (Method) $\times$ 3 (Video) ANOVA. F values are reported for significant effects; Greenhouse–Geisser corrections were applied, and p values for the SSQ outcomes are Holm-corrected across the four outcomes. Time: $df = 1, 35$.

Effect	TSC	Nausea	Oculomotor	Disorientation	Presence
Time	$F = 29.15$, $p < .001$	$F = 11.08$, $p = .002$	$F = 23.97$, $p < .001$	$F = 28.64$, $p < .001$	–
Method	$p = 1.000$	$p = 1.000$	$p = 1.000$	$p = 1.000$	$p = .758$
Video	$p = .101$	$p = .253$	$F = 5.72$, $p = .021$	$p = .192$	$p = .311$
Time $\times$ Method	$p = 1.000$	$p = 1.000$	$p = 1.000$	$p = 1.000$	–
Time $\times$ Video	$F = 7.73$, $p = .004$	$p = .060$	$F = 9.61$, $p = .001$	$F = 6.91$, $p = .008$	–
Method $\times$ Video	$p = 1.000$	$p = 1.000$	$p = 1.000$	$p = 1.000$	$p = .568$
Time $\times$ Method $\times$ Video	$p = 1.000$	$p = 1.000$	$p = 1.000$	$p = 1.000$	–

Table 7: Main quantitative results of Study 2 by method, collapsed across video type (participant means $\pm$ SD, $n = 36$). Δ is the post-minus-pre change; positive values mean symptoms increased during viewing.

Method	Pre TSC	Post TSC	ΔTSC	ΔNausea	ΔOculomotor	ΔDisorientation	Presence
ROF	9.90 ± 18.65	18.91 ± 21.18	9.00 ± 9.56	3.97 ± 6.24	7.65 ± 9.86	13.66 ± 15.45	5.10 ± 1.04
Minimap	9.38 ± 15.02	18.46 ± 20.51	9.07 ± 11.39	3.27 ± 9.33	8.91 ± 11.83	12.63 ± 16.32	5.12 ± 1.05
Overlay Path	9.35 ± 19.45	16.59 ± 21.59	7.24 ± 12.60	2.56 ± 7.93	6.67 ± 12.28	10.96 ± 17.22	5.04 ± 1.06

Table 8: Non-inferiority of the trajectory-based interfaces. Difference in TSC points (trajectory minus comparison; positive = more sickness). Non-inferior when the upper limit of the 90% CI is below the $+5$ margin. Study 2 uses the pre-to-post change; Study 1 uses post-trial TSC.

Comparison	Difference	90% CI	Non-inferior
Study 2 (vs ROF)			
Trajectory (pooled)	-0.85	$[-2.54, 0.84]$	**yes**
Minimap	$+0.07$	$[-2.57, 2.71]$	**yes**
Overlay Path	-1.77	$[-4.22, 0.69]$	**yes**
Study 1 (vs Dynamic FOV)			
Trajectory (pooled)	$+7.81$	[0.54, 15.08]	no
Minimap	$+7.20$	[0.16, 14.25]	no
Overlay Path	$+8.42$	$[-0.08, 16.91]$	no

Comparison with Study 1. Applying the same contrast to Study 1, with Dynamic FOV as the non-trajectory comparison, did not establish non-inferiority: post-trial TSC was 7.8 points higher with the trajectory-based interfaces than with Dynamic FOV (95% CI $[-0.9, 16.5]$). This interval includes both zero and the margin, so the comparison is inconclusive. In Study 1, the trajectory-based interfaces still reduced TSC by 15.5 points relative to Baseline (95% CI [6.5, 24.4]) and produced higher Presence than Dynamic FOV ($+0.55$, 95% CI [0.26, 0.84]).

5.5.3 Presence

Unlike in Study 1, **Presence did not differ between methods** ($F(1.84, 64.48) = 0.25$, $p = .758$); the effects of Video and Method $\times$ Video were also not significant. The trajectory-based interfaces scored 0.02 points lower than ROF (95% CI $[-0.23, 0.19]$), and all three methods averaged about 5.1 on the 7-point scale. Presence was unrelated to the increase in sickness ($r = -.04$, $p = .82$). None of the three methods therefore showed the Presence cost that Dynamic FOV showed in Study 1.

5.5.4 Intrusiveness-Related Complaints

The trajectory-based interfaces did not receive fewer recorded complaints than ROF (Figure 6). Thirteen participants (36%) complained about ROF, 7 (19%) about Minimap, and 13 (36%) about Overlay Path. The mean within-participant difference was $+0.08$ (bootstrap 95% CI $[-0.08, 0.26]$; exact one-sided $p = .247$), and Minimap alone did not differ significantly from ROF (exact McNemar test, Holm-corrected $p = .36$). Instead, the complaints differed in *kind*:

ROF. Flicker and visual clutter were ROF's main problem (9 participants, compared with 1 for Minimap and none for Overlay Path), e.g., "too many motion vectors" or white streaks that caused glare. Opinions on occlusion were split (7 for, 7 against). At the same time, ROF was the method most often praised for realism (13), for example for creating a "wind effect" or blending with the surrounding scene.

Minimap. No participant said that the Minimap blocked the scene, and 13 praised it for keeping the view clear; 10 valued its preview of the route. Attention was the main issue: 5 participants found it distracting, while 5 others said they did not notice it or forgot to look at it.

Overlay Path. Anticipating upcoming turns was the most frequently cited advantage of any method in the study (17 participants), and 11 found the Overlay Path realistic. However, it also received the most distraction complaints (8), and 6 participants said that it was too large or blocked the scene.

The comparison depended on which complaint types were counted. Excluding distraction, where Overlay Path received the most complaints, produced a significant difference ($p = .012$), whereas excluding visual noise, ROF's main problem, removed any difference ($p = .70$). The interviews were coded by one researcher and inter-rater reliability has not been assessed, so we treat this analysis as exploratory.

5.5.5 Preference Ranking

Unlike in Study 1, all 36 participants provided a complete ranking (Table 9). **No method was preferred overall**: the rankings did not differ (Friedman $\chi^2(2) = 0.72$, $p = .697$), and participants barely agreed with one another (Kendall's $W = .010$). ROF was ranked last by 16 participants (44%, 95% CI [28, 62]), which was not more often than expected by chance ($p = .11$). In the Plackett–Luce model, each trajectory-based interface had an estimated probability of .36 of being ranked first, compared with .27 for ROF, but this difference was not significant (worth ratio 1.34, 95% CI [0.78, 2.32], $p = .28$).

The secondary questions showed the same picture. Of the 21 participants who named the method that best reduced their sickness, 8 chose ROF, 7 Minimap, and 6 Overlay Path; of the 26 who named the most realistic method, 11 chose ROF, 7 Minimap, and 8 Overlay Path.

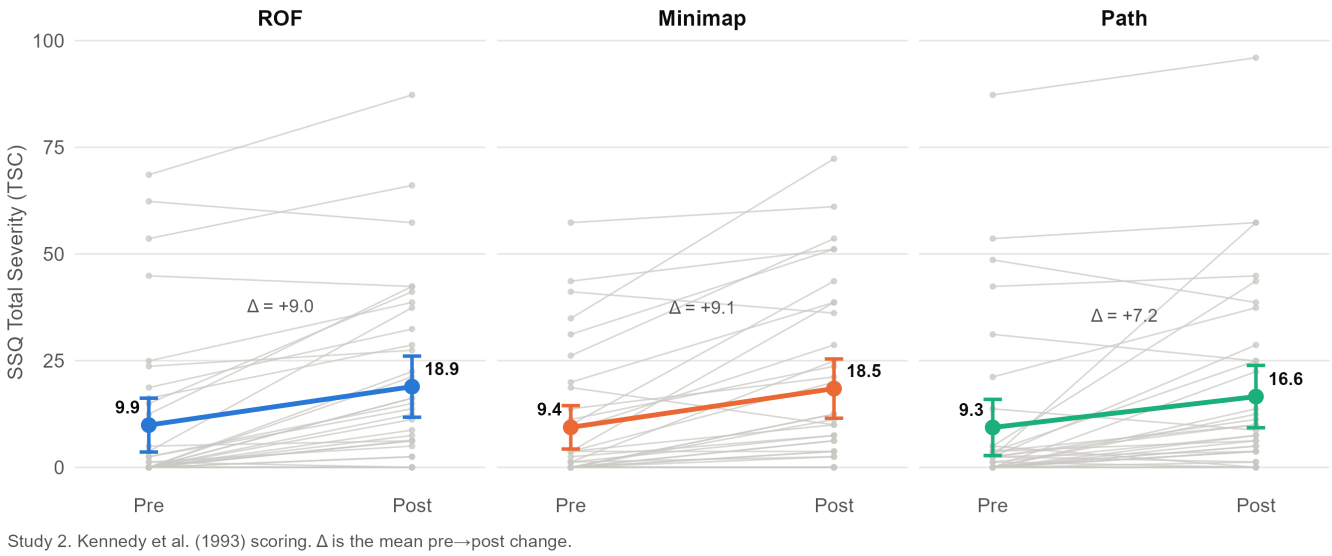


Figure 4: SSQ Total Severity before and after each video in Study 2. Grey lines are individual participants (means over the three videos); colored points show the mean with 95% CI. Sickness rose by a similar amount under all three methods.

Table 9: Preference ranking in Study 2 ($n = 36$). Lower mean rank indicates stronger preference. Borda score: 2 points for first place, 1 for second.

Method	Mean rank	Borda	First	Last
Overlay Path	**1.92**	**39**	**13**	**10**
Minimap	1.97	37	11	**10**
ROF	2.11	32	12	16

When explaining their first choice, 86% of participants referred to interface experience, such as occlusion, distraction, or route anticipation, whereas only 19% referred to sickness and 25% to realism. **Participants' sense of "balance" therefore rested mainly on how each interface fit into the viewing experience**, rather than on perceived sickness reduction.

5.5.6 Study 2 Summary

Table 10 combines the quantitative and qualitative results. Study 2 shows that **the trajectory-based interfaces were non-inferior to ROF in measured VIMS**: sickness increased by a similar amount under all three methods, and the data rule out the trajectory-based interfaces being more than 0.84 TSC points worse than ROF on average. **Presence was equally high for all three methods.** The data did *not* show that the trajectory-based interfaces were less intrusive: they drew about as many recorded complaints as ROF, but of a different kind. ROF was criticized for flicker, Overlay Path for distraction and occlusion, and Minimap drew the fewest complaints. Participants did not agree on a single most balanced method; their rankings followed these trade-offs rather than perceived sickness reduction.

These results have three limitations. First, Study 2 had no Baseline, so it cannot show that any method reduced VIMS relative to unmodified video. Second, sickness was mild (post-trial TSC 16.6–18.9, compared with 48.3–71.6 in Study 1), which may have left little room for the methods to differ. Third, the complaint analysis relies on unprompted comments coded by one researcher.

6 Discussion

7 Conclusion

References

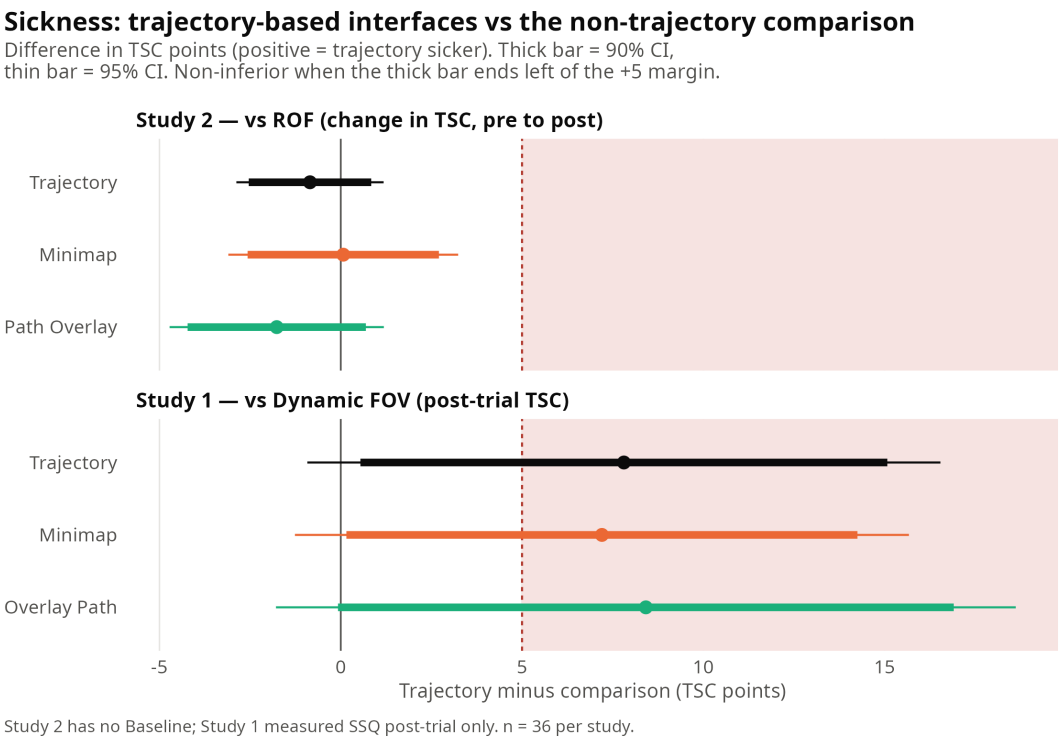


Figure 5: Sickness difference between the trajectory-based interfaces and the non-trajectory comparison in each study. Thick bars show the 90% CI and thin bars the 95% CI; the dashed line marks the $+5$ TSC non-inferiority margin.

Table 10: Integration of quantitative and qualitative findings in Study 2. Interview counts are numbers of participants (out of 36).

Method	**Quantitative result**	**Interview feedback**
Minimap	TSC increased by 9.07; non-inferior to ROF (upper limit 2.71). Presence 5.12.	Fewest intrusiveness-related complaints (7). Never described as blocking the scene; route preview valued (10). Some found it distracting (5) or forgot to look at it (5).
Overlay Path	Numerically smallest TSC increase (7.24); non-inferior to ROF (upper limit 0.69). Presence 5.04.	Most-cited advantage in the study: anticipating turns (17); seen as realistic (11). Most distraction complaints (8); too large or blocking the scene (6).
ROF	TSC increased by 9.00. Presence 5.10.	Most often praised for realism (13), e.g., a “wind effect”. Most complaints about flicker and visual clutter (9). Ranked last most often (16).

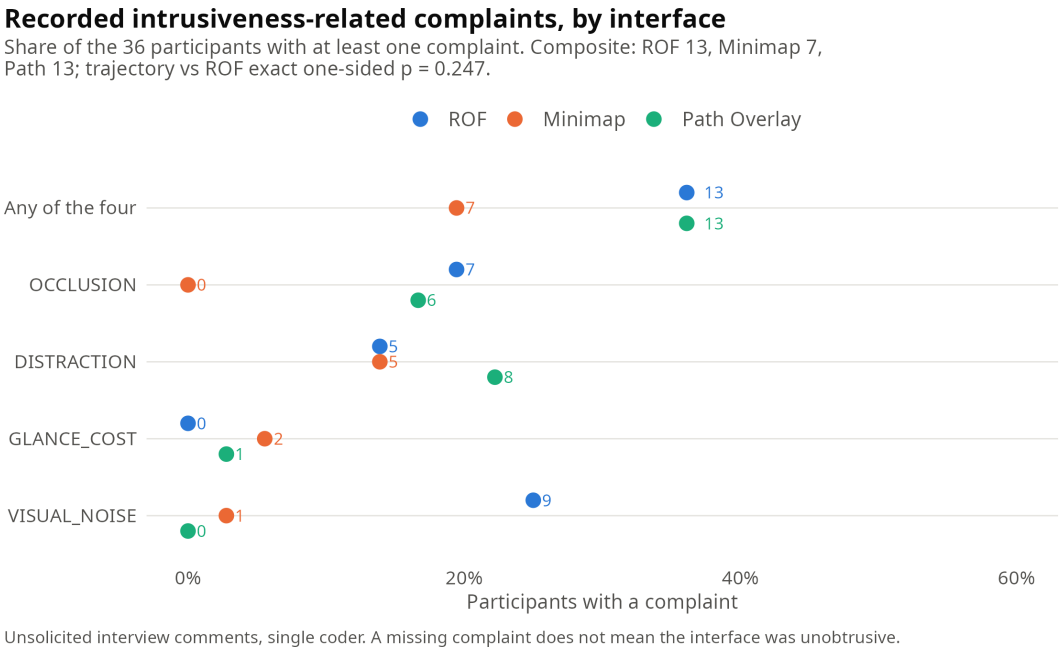


Figure 6: Participants who made at least one recorded intrusiveness-related complaint about each method, overall (“any of the four”) and by complaint type. Counts are out of 36 participants.